Jam processing and storage effects on blueberry polyphenolics and antioxidant capacity.

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Fresh blueberries were processed into sugar and sugar-free jams and stored for 6 months at 4 and 25 degrees C. The jams were analyzed immediately after processing and over 6 months of storage for polyphenolic content, percent polymeric color, and antioxidant capacity. Processing resulted in losses of anthocyanins, procyanidins, chlorogenic acid, and ORAC in both jam types, but flavonols were well retained. Marked losses of anthocyanins and procyanidins occurred over 6 months of storage and were accompanied by increased polymeric color values. Chlorogenic acid levels also declined during storage, but flavonols and ORAC changed little. Jams stored at 4 degrees C retained higher levels of anthocyanins, procyanidins, and ORAC and had lower polymeric color values than jams stored at 25 degrees C. Sugar-free jams retained higher levels of anthocyanins and had lower polymeric color values than sugar jams late during storage. Blueberry jams should be refrigerated to better retain polyphenolics and antioxidant capacity.